Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/01/2026
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Project Title:	Temporal Impact of Simulated Microgravity on Ocular Vascular Hydrodynamics		
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Key Personnel Changes/Previous PI:	NONE		
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Task Description:	Spaceflight associated neuro-ocular syndrome (SANS) is reported to affect about 40% of astronauts completing long-duration spaceflights and is characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, retinal nerve fiber layer thickening, or choroidal folds. Although prolonged exposure to the headward fluid shift occurring in microgravity is regarded as the primary instigating factor, the pathophysiology of SANS remains unclear. An additional gap in our knowledge is whether ground-based models of microgravity can simulate SANS. Therefore, ground-based studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we propose to focus on potential changes directly to the ocular vasculature caused by simulated microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen is dependent on retinal blood flow. Recent studies have reported elevated central retinal artery blood flow velocity during long-term spaceflight. A sustained increase in retinal blood flow during microgravity exposure may contribute to optic disc edema because of the elevated transcapillary pressure. In addition, a microgravity induced headward venous fluid shift, resulting in elevated cephalad venous pressures, may impede removal of ocular interstitial fluid, resulting in optic disc edema and contribute to SANS. The development of optic disc edema also may be exacerbated by disruption of the blood-retinal barrier and/or reduction in ocular lymph flow. A potential molecular mechanism promoting these vascular changes in the eye is an inflammatory response related to edema formation. Elevated tissue fluid increases the diffusion distance for oxygen and other nutrients. Edema also compromises diffusion mediated removal of toxic cellular metabolites. However, so far there has been no systematic analysis of ocular vascular changes in microgravity that may promote ocular edema and/or other characteristics of SANS. To address these information gaps, we have assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics). The objective of this application is to determine whether the hindlimb unloading (HU) model of simulated microgravity causes a time-dependent change in structure and function of the ocular vasculature at the level of feed arteries, venous exchange vessels and lymph vessels, and whether sex differences influence these responses. We will perform in vivo measures of vascular function (retinal artery blood flow velocity, retinal arteriolar and venular diameters, and retinal venular permeability) and in vitro studies of isolated vessel structure and function (vessel histology, arterial vasomotor regulation, and lymphatic transport characteristics) in male and female mice after 2- and 4-wk of HU. The central hypothesis of this proposal is that simulated microgravity causes temporal changes in structure/function of the ocular vasculature and subsequent alterations in ocular hydrodynamics leading to SANS. We will pursue 3 specific aims: (1) Determine the temporal impact of simulated microgravity on ocular artery structure/function; (2) Determine the temporal impact of simulated microgravity on ocular vein structure/function; (3) Determine the temporal impact of simulated microgravity on ocular lymphatic structure/function. Our team approach to accomplish these studies will provide a novel comprehensive evaluation of the temporal impact of simulated microgravity on ocular hydrodynamics in mice, and correlation of these findings with SANS. Outcomes will help identify the strengths and limitations of a murine HU model for SANS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our ground-based studies in this project will address whether relatively short- and long-term simulated microgravity using the hindlimb unloading model in mice can simulate signs and symptoms of spaceflight associated neuro-ocular syndrome (SANS), such as ocular edema and thickening of the retinal tissue. We will also determine whether simulated microgravity causes a time-dependent change in structure and function of the blood vessels in the eye, and whether ocular vascular changes correlate with SANS. Results from this project are expected to provide the first insight into the impact of simulated microgravity on vascular hydrodynamics in the eye. The expected findings will likely also provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma and ocular hypertension.
Task Progress:	A major complication of astronauts during and following long-duration spaceflight is the development of disturbances in vision, along with swelling and accumulation of fluid in the retinal tissue of the eye. NASA has termed these visual disturbances, "Spaceflight Associated Neuro-ocular Syndrome", or SANS. Because astronauts in space are exposed to microgravity, their bodies respond to fluid shifts from the lower to upper body and the lack of gravitational loading. The vision changes in astronauts are thought to occur at least in part by this adaptation in space and changes in the circulatory system. However, so far there has been no detailed analysis of blood vessel changes in the eye during microgravity that may promote swelling and vision complications. Therefore, we characterized the three components of the circulatory system (arteries, veins, and lymphatics) in the eye in response to simulated microgravity in an animal model. The goal of this study was to determine whether simulated microgravity causes a time-dependent change in structure and function of the blood vessels in the eye – at the level of arteries, veins, and lymph vessels – and whether both males and females are affected. We also determined whether blood vessel changes correlate with fluid accumulation and thickening of the vision-sensing neural cell layer in the retina. The impact of simulated microgravity for 2 weeks and 4 weeks on ocular vessel function in male and female animal subjects was investigated. For ocular artery studies, the ability of these vessels to widen or dilate was slightly reduced after 4 weeks of simulated microgravity in male and female subjects. By contrast, the ability of the ocular arteries to narrow or constrict was diminished in female, but not male, subjects after 4 weeks of simulated microgravity. Outcomes with veins revealed that simulated microgravity for 4 weeks did not alter the leakiness of the small retinal veins in the eye, and this was consistent with the maintenance of the integrity of the structural proteins in the small retinal veins. For lymphatic vessel studies, simulated microgravity did not produce a consistent alteration in ocular lymphatic function – either from the tonic contraction or phasic lymph pumping perspective – in male and female subjects. It also appears that simulated microgravity did not promote a significant amount of fluid accumulation in the retina around the optic nerve in both male and female subjects. Overall, these results suggest that simulated microgravity for 4 weeks has a slight impact on the function of ocular arteries, but inconsistent effects on ocular veins and lymphatics in male and female subjects. The outcomes do not provide support for a ground-based animal model of SANS using a duration of 4 weeks of simulated microgravity to study its impact on ocular vascular function. Potential considerations to extend these findings would be to increase the duration of simulated microgravity, and to expose subjects to multiple space environment factors (such as simulated microgravity with space radiation, and/or elevated carbon dioxide levels).

Bibliography Type:	Description: (Last Updated: 07/07/2026)